Paleomagnetism of Late Cretaceous Granitic Plutons in the Sierra Nevada, California: Further Results

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Abstract. Additional sampling in another area and re-evaluation of previously rejected data have provided paleomagnetic poles for five new sites in the 85-million-year-old Cretaceous granites of the Sierra Nevada, California. The mean of these five poles and the nine poles previously reported is located at 165°W and 69°N, with the semiangle of the 95% confidence cone equal to 9.6°. This mean paleomagnetic pole differs only slightly from that previously reported for nine sites only and agrees with two Cretaceous poles from rocks in Quebec; thus it may represent a geographic pole as well. There is no systematic distribution of individual virtual poles with respect to their mean. The only ferromagnetic mineral in all these rocks is pure magnetite, which probably accounts for the large observed dispersion of NRM directions and the magnetic instability of many of the specimens. Measurements of susceptibility anisotropy indicate that only about half of the dispersion can be caused by random anisotropy and that the observed uniform anisotropy at some sites has not biased the NRM directions significantly.

Introduction. The remanent magnetization of granitic rocks in the vicinity of Yosemite Valley, California, which constitute a part of the main composite batholith of the Sierra Nevada, has been described by *Curie et al.* [1963]. The present paper gives further results from some of the previously discarded sampling sites in that area, together with new data from similar rocks exposed along the Sonora Pass highway, some 40 miles north of Yosemite Valley. These granitic plutons are all Late Cretaceous in age. Paleomagnetic results from the Late Jurassic Valley Series batholiths of basic to intermediate composition which occur along the western margin of the Sierra Nevada will be described in a future paper.

The purposes of the present study have been (1) to confirm the paleomagnetic pole position given by Curie et al., (2) to determine the significance of the peculiar distribution of virtual poles around the mean pole described in that paper, and (3) to investigate more thoroughly the stability and origin of the remanent magnetization observed in these rocks. Unless otherwise indicated, experimental methods are the same as in the earlier work.

Yosemite Valley area. Less than half of the oriented cores collected in the region of Yosemite Valley in 1961 and 1962 were considered by Curie et al. to possess a reliable record of paleomagnetic field direction. Data were accepted or rejected on a site-by-site basis, the primary criterion being the degree of dispersion of directions for each site as a whole. In most cases it was found that excessive scatter could be correlated with such characteristics as location of the site on a natural rather than artificial exposure, actual instability of remanent magnetization, extreme variation in intensity of magnetization, or planar distribution of magnetization directions. In retrospect, it seems that these considerations might be applied in part to individual specimens within a site and that some of the previously rejected data can be considered reliable.

Magnetization directions of all specimens from three such sites are shown in Figure 1, and Table 1 summarizes the data from the specimens considered acceptable. The locations of these sites may be found in Figure 1 of *Curie et al.* [1963], where they were designated as 'rejected sampling sites,' along with eight other sites which yielded completely unusable data. The geologic relationships and K-Ar ages of the various plutons and a description of the autoliths are also given in that paper. A brief discussion of the three sites follows.

Site A′, in Half Dome quartz monzonite, is on

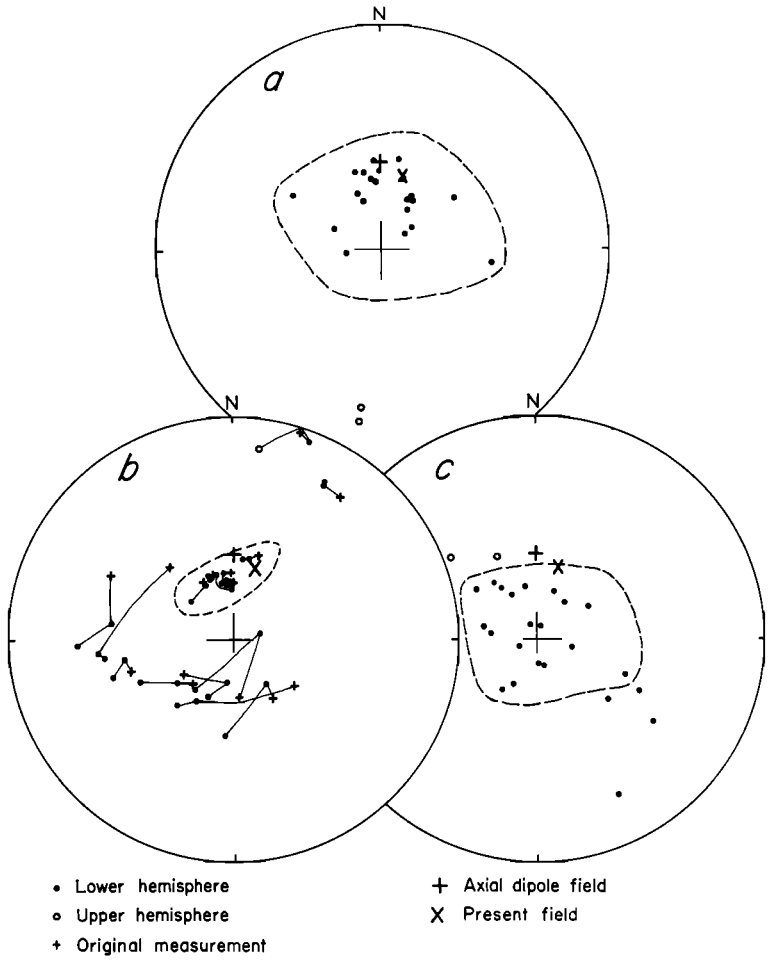


Fig. 1. Equal-area projections of north-seeking magnetization directions at sites in the Yosemite area. (*a*) Site A′. (*b*) Site F′. (*c*) Site G′. In (*b*), lines connect original and successive repeat measurements on the same specimens. Dashed lines enclose directions used to calculate means given in Table 1.

a natural (glaciated) exposure located at the west end of Tenaya Lake between sites A and B. Magnetization directions as measured 18 months after collection are shown in Figure 1*a*, in which it is seen that only two specimens, cut from a single core, are widely divergent from the rest. These possess anomalously intense NRM, about 5×10^{-2} emu/cc. Of the remaining specimens, 18 are from autoliths and 3 are from the enclosing quartz monzonite. Table 1 shows that the dispersion decreased significantly during storage in the earth's field in the laboratory but that the movement of the mean direction is apparently unrelated to any influence the earth's present field may have had on the specimens before collection.

Site F′, in El Capitan granodiorite, is in an artificial exposure on the Glacier Point road, approximately 1 mile south of site F. NRM directions from this site, as measured immediately after collection, 1 month afterward, and 18 months afterward are shown in Figure 1*b*. (The first two sets of measurements were previously published in Figure 2*a* of Curie et al.) It is clear that the NRM directions fall into two groups: one possesses little scatter and good stability; the other is unstable and widely scattered. This grouping coincides with the physical

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TABLE 1. Mean NRM Directions of Acceptable Specimens from Three Additional Sites, Yosemite Valley Area
(*a:* First measured, June 1962. *b:* Remeasured, July 1962. *c:* Remeasured, January 1964.)

Site	N		k	α_{95}	D	I	$J_n \times 10^4$
A'	21	*a*	10.7	10.2	16	68	8.3
		c	19.8	7.3	4	71	10.0
F'	6	*a*	102	6.7	−5	66	3.0
		b	133	5.8	−9	66	3.1
		c	100	6.7	−15	67	3.1
G'	19	*a*	13.9	9.3	−39	79	1.2
		c	18.2	8.1	−28	85	1.2

Symbols in all tables are as follows: N is the number of specimens; k is Fisher's precision parameter, and α_{95} is the semiangle of the 95% confidence interval centered on a mean direction [*Fisher*, 1953]; δ_p and δ_m are the semiaxes of the 95% confidence oval centered on a virtual geomagnetic pole, respectively parallel and perpendicular to the paleomagnetic meridian [*Cox and Doell*, 1960]; D is the mean declination, positive east of north; I is the mean inclination, positive downward; J_n is the intensity of NRM in emu/cc; χ is the reversible volume susceptibility.

location of the cores on the outcrop; the stable specimens are from three cores spaced about a meter apart in a very large autolith, and the unstable specimens came from smaller autoliths and host granite. Although the most recent measurements give a somewhat lower value of k than the previous ones, they are considered the most reliable because the consistent increase in westerly declination suggests decay of viscous magnetization previously acquired in situ (Table 1).

Site G′, in Sentinel granodiorite, is in an artificial exposure on the Glacier Point road, about 1/4 mile west of site G. The magnetization directions measured 18 months after collection are shown in Figure 1*c*. In this case the criteria for rejection of directions were either divergence from the apparent mean or instability, and all but one of the unusable specimens were located in a single restricted group at one end of the sampling site. Of the usable specimens, ten were of grandiorite and nine of autolith material. As with site A′, Table 1 shows that the dispersion of the NRM directions decreased signficantly after storage for several months.

The decreases in dispersion of NRM during laboratory storage of the specimens from sites A′ and G′ is probably due mostly to decay of viscous magnetization, which varies in its initial intensity from specimen to specmen. Also, because these granites possess a component of relatively soft magnetization, there is a possibility that some specimens may have been very slightly remagnetized during coring and trimming. Spontaneous decay of such randomly directed IRM would contribute to the decrease of the dispersion, but the effect must be small.

The virtual geomagnetic south poles with semiaxes of the associated 95% confidence ovals are given in Table 2, calculated from the mean magnetization directions at the three sites described above and the nine sites, A through J, from Curie et al. For all sites, the most recent set of measurements has been used, rather than the set giving highest k as had been done previously, partly for the sake of consistency but mainly in order to minimize any systematic bias due to viscous magnetization. It is noteworthy that in Table 5 of Curie et al. for seven of the nine sites the more recent measurements gave a mean direction having a more westerly declination, i.e. farther from the present field direction.

All the virtual site-poles in Table 2 are plotted in Figure 2 with their ovals of 95% confidence. The mean of these poles is also shown; the addi-

TABLE 2. Virtual Geomagnetic South Poles and Mean Pole from Yosemite Valley Region
All data from specimens remeasured after storage, except those from site B which were demagnetized in 10 oe. Sites A through J (unprimed) from data in *Currie et al.* [1963].

Site	N	W. Long.	N. Lat.	δ_p	δ_m
A	15	179	57	6.5	8 1
A′	21	111	73	11.0	12.7
B	21	178	77	8.4	10.8
C	21	184	80	7.0	9.1
D	6	219	62	6.8	10.4
E	8	156	84	8.3	10.8
F	9	230	82	7.9	11.2
F′	6	156	74	9.3	11.2
G	26	175	55	6.9	8.3
G′	19	127	47	15.8	16.0
H	9	191	44	6.4	8.7
J	20	130	84	9.8	12.5
Mean*	12	171	71		

* $k = 19.2$; $\alpha_{95} = 10.2$.

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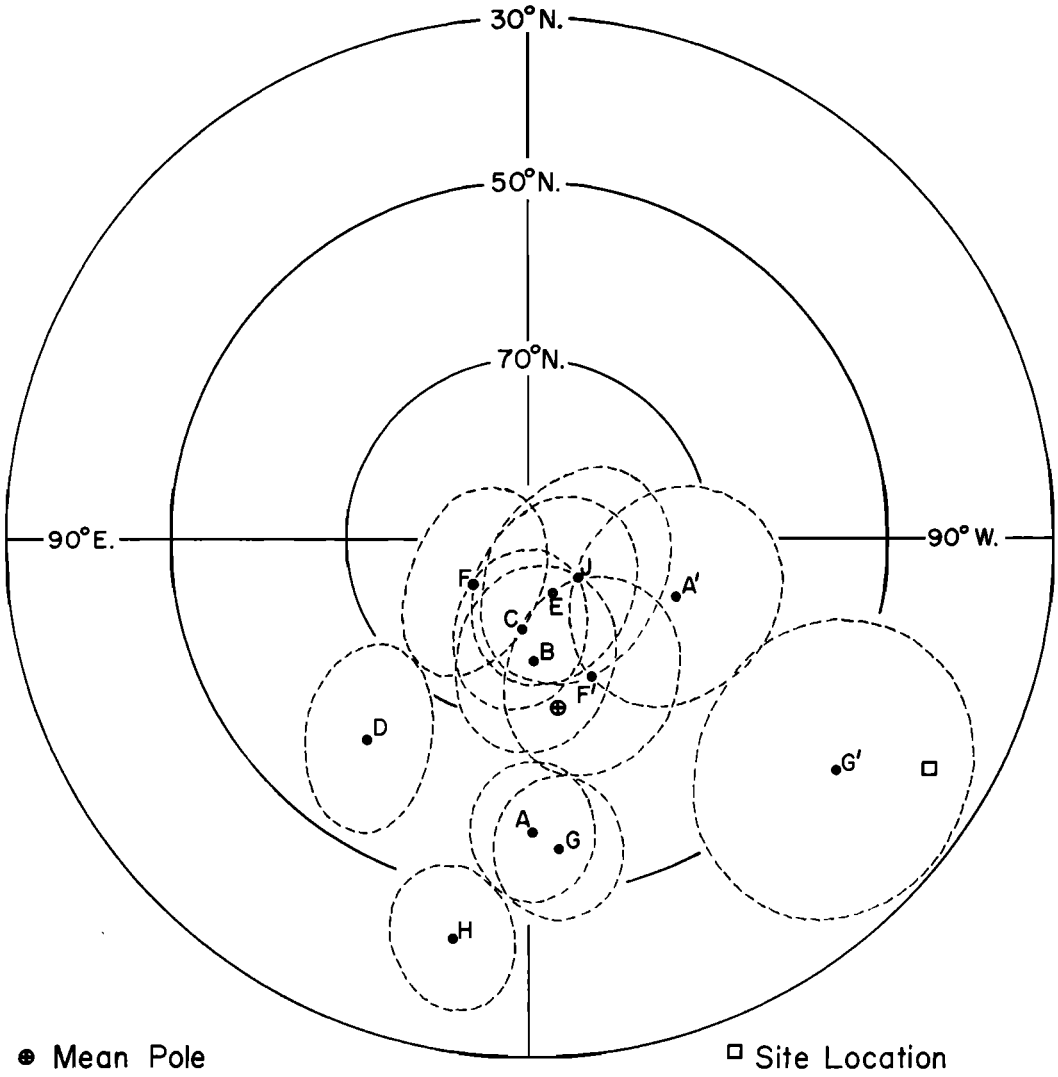


Fig. 2. Equal-area map of northern hemisphere above latitude 30° showing all virtual geomagnetic south poles with 95% confidence ovals, from the Yosemite area (Table 2). Poles A through H are from *Curie et al.* [1963], poles A′, F′, G′ represent new data.

tion of data from three more sites has not significantly changed its position from that previously given for nine sites by Curie et al. It is also evident from Figure 2 that the elliptical distribution of poles approximately equidistant from the mean pole that was found by them has not been substantiated by the addition of more data from the same region and thus must have been a fortuitous result of limited sampling.

Sonora Pass area. The plutons of the Yosemite Valley area constitute a relatively small part of the main Sierra Nevada batholith, and it was thought advisable to extend the sampling to another area. Accordingly, cores were taken at five sites along California State Highway 108, which crosses the Sierra at Sonora Pass. The locations of four of these sites are shown in Figure 3, together with what little is known of the bedrock geology of this area. This map is partly from *Slemmons* [1953], who gave brief petrographic descriptions of the rocks. The Topaz Lake quartz monzonite contains abundant large K-feldspar phenocrysts and few autoliths, is the youngest pluton in this area, and underlies the

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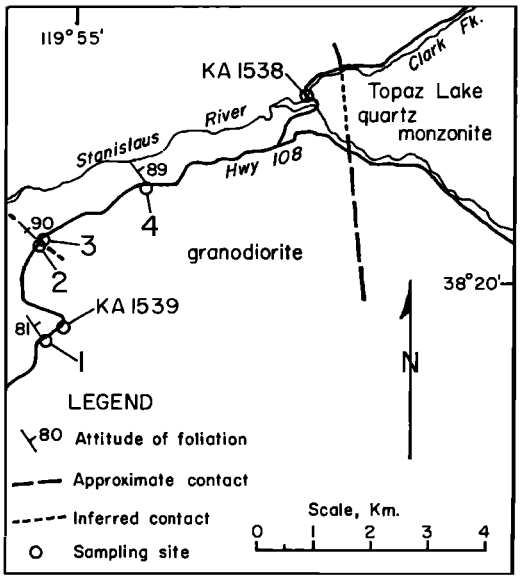


Fig. 3. Geologic sketch map of Dardanelles area, west of Sonora Pass, showing relations of granitic plutons and sampling localities.

crest of the Sierra at this latitude. It is thus equivalent to the Cathedral Peak quartz monzonite of the Yosemite area (cf. Figure 1 of *Curie et al.* [1963]) and is probably about 80 m.y. old. The granodiorite immediately to the west occupies a position analogous to the Half Dome quartz monzonite or the Sentinel granodiorite of the Yosemite area. During the present work a vertical dike-like strongly foliated zone containing up to 80% deformed autoliths was mapped between sites 2 and 3 (Figure 3), which appears to be a contact between two granodiorite bodies. Through the courtesy of Professor J. F. Evernden, K-Ar ages have been determined for biotite and hornblende concentrates from granodiorite samples obtained at two localities shown in Figure 3. The analytical data and ages are given in Table 3; there is good agreement between replicate argon analyses on biotite and between ages of biotite and hornblende from the same locality. The biotite ages cover the same range as biotite ages from the Half Dome, Sentinel, and El Capitan plutons in the Yosemite area [*Curie et al.*, 1963; *Evernden et al.*, 1957]. The westernmost granodiorite is clearly older, reinforcing the interpretation of the zone between sites 2 and 3 as a contact between plutons, even though the two granodiorites are petrographically similar.

The granodiorite at sites 1–4 is noticeably foliated; generalized attitudes are shown in Figure 3. The foliation is barely visible in the granodiorite itself but is clearly defined by extreme flattening of autoliths, some of which have dimension ratios as much as 15 : 1, in contrast to the equant autoliths commonly found elsewhere. The possible paleomagnetic significance of this foliation is discussed in a later section.

The fifth sampling site in the Sonora Pass area is on the eastern side of the mountain range, about 10 km east of sites 1–4. It is located on a glaciated outcrop of Topaz Lake quartz monzonite, near the eastern margin of the pluton, in an area where large K-feldspar phenocrysts and autoliths are abundant. The autoliths are round or oval, and little or no foliation is evident.

Mean magnetization directions and statistical parameters for all five sites in the Sonora Pass area are given in Table 4 and Figure 4*a*. (Note that in Figure 4 only the central 50° of the reference spheres are shown in the projections.) Except for site 5, the mean NRM directions plotted in Figure 4*a*, which are derived from the initial set of measurements, are clearly anomalous with respect to the results from the Yosemite area. However, nearly all specimens proved to carry a large component of unstable magnetization. All specimens were measured as soon as possible after collection, and again 7 months later. During this period the cores were stored

TABLE 3. Potassium-Argon Ages of Mineral Concentrates from Sonora Pass Granodiorite (J. F. Evernden, 1964, personal communication)
Decay constants: $\lambda_\epsilon = 0.585 \times 10^{-10}$ yr^{-1}; $\lambda_\beta = 4.72 \times 10^{-10}$ yr^{-1}. Sample localities shown in Figure 3.

Sample	Mineral	Sample wt., g	K, wt. %	Ar^{40}_{at}, %	Age, m.y.
KA 1538	biotite	0.3425	7.591	5	83.0
KA 1538	biotite	0.4074	7.591	8	83.6
KA 1538	hornblende	2.030	0.5848	23	84.2
KA 1539	biotite	0.4743	7.834	8	87.0
KA 1539	hornblende	2.025	0.8207	13	90.1

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TABLE 4. Mean NRM Directions of All Specimens from Sonora Pass Area (*a*: First measured, August 1963. *b*: Remeasured, February 1964.)

Site	N		k	α_{95}	D	I
1	29	*a*	11.1	8.4	56	80
		b	4.4	14.5	−4	65
2	15	*a*	65	4.8	78	65
		b	1.9			
3	20	*a*	28	6 2	177	88
		b	6.5	13.9	−39	75
4	52	*a*	12.5	5.8	37	86
		b	7.4	7.8	−15	68
5	30	*a*	21	5.8	−20	72
		b	5.7	12.1	−16	58

inverted in the earth's field; thus their NRM vectors made, on the average, obtuse angles with the field in the laboratory, and any remagnetization would be expected to cause more conspicuous changes in direction than if the cores had been left upright. Table 4 shows that in most of these specimens there is a large component of viscous magnetization with a relatively short time constant; for all five sites the dispersion increased drastically during the storage period. Note especially site 2, for which the precision parameter k decreased from the respectable value of 65 to 1.9, which represents a random distribution. In view of this, it is difficult to understand why the NRM directions as originally measured gave fairly tight groupings which were nevertheless significantly different from both the present field and the axial dipole field directions and why no appearance of planar distributions of NRM directions was observed for any of the sites.

It remains to account for the difference in magnetic behavior of the Sonora Pass and Yosemite granites, and to try to recover some useful paleomagnetic information from the Sonora Pass sites. Table 5 gives the average intensities of NRM and reversible susceptibilities of all granite and autolith specimens from the eight sites described above; it may be compared with Table 3 of *Curie et al.* [1963]. There are no significant differences in susceptibilities between the Sonora Pass and Yosemite rocks, nor are there any in NRM intensities, with the obvious exception of site 4. The distributions of intensities of NRM in granites and autoliths at all Sonora Pass sites grouped together are shown in Figure 5. None of the granite specimens has anomalously high intensity, and most of the strongly magnetized autolith cores are from site 4, accounting for its high average intensity.

Such strongly magnetized autoliths occur at sites 1, 3, and 4; it turns out that their magnetization is relatively much more stable than that of the other specimens. The behavior of the NRM of the 24 specimens (of autoliths), having intensity of magnetization greater than 7×10^{-4} emu/cc when first measured, is shown in Table 6 and Figure 4*b, c, d*. Changes in mean direction for all three sites were minimal during storage and were not excessive during ac demagnetization in 200 oe peak field. Some specimens were stable to 400 oe. For site 3, demagnetization caused a significant increase in dispersion, but, since there are only two strongly magnetized specimens from this site, it carries insufficient weight for calculation of a virtual pole. The stability and uniformity of NRM directions in the strongly magnetized specimens from these three sites cannot be due to lightning or any artificial event, and they are significantly different from the axial dipole and present field directions; thus we conclude that the mean directions after demagnetization for sites 1 and 4 as given in Table 6 represent paleomagnetic field directions. Although their circles of confidence overlap, they are probably different in age, and the two corresponding virtual pole positions are given in Table 6.

Dispersion of NRM directions. It has been assumed throughout this work that the NRM directions at each site are distributed according to *Fisher's* [1953] distribution. In the previous paper [*Curie et al.*, 1963] it was stated that, based on results of ac demagnetization, the dispersion appears due to a randomly directed component of nearly constant intensity superimposed on the directed component. *Cox* [1964] has shown that this circumstance leads to a non-Fisherian distribution in which most of the vectors fall near the margin of the group, but clearly none of the sets of NRM vectors from the Sierran granites follow this kind of distribution (cf. Figure 1, for example). It appears more likely that there is a distribution of intensities of random components among specimens, corresponding to the second model dis-

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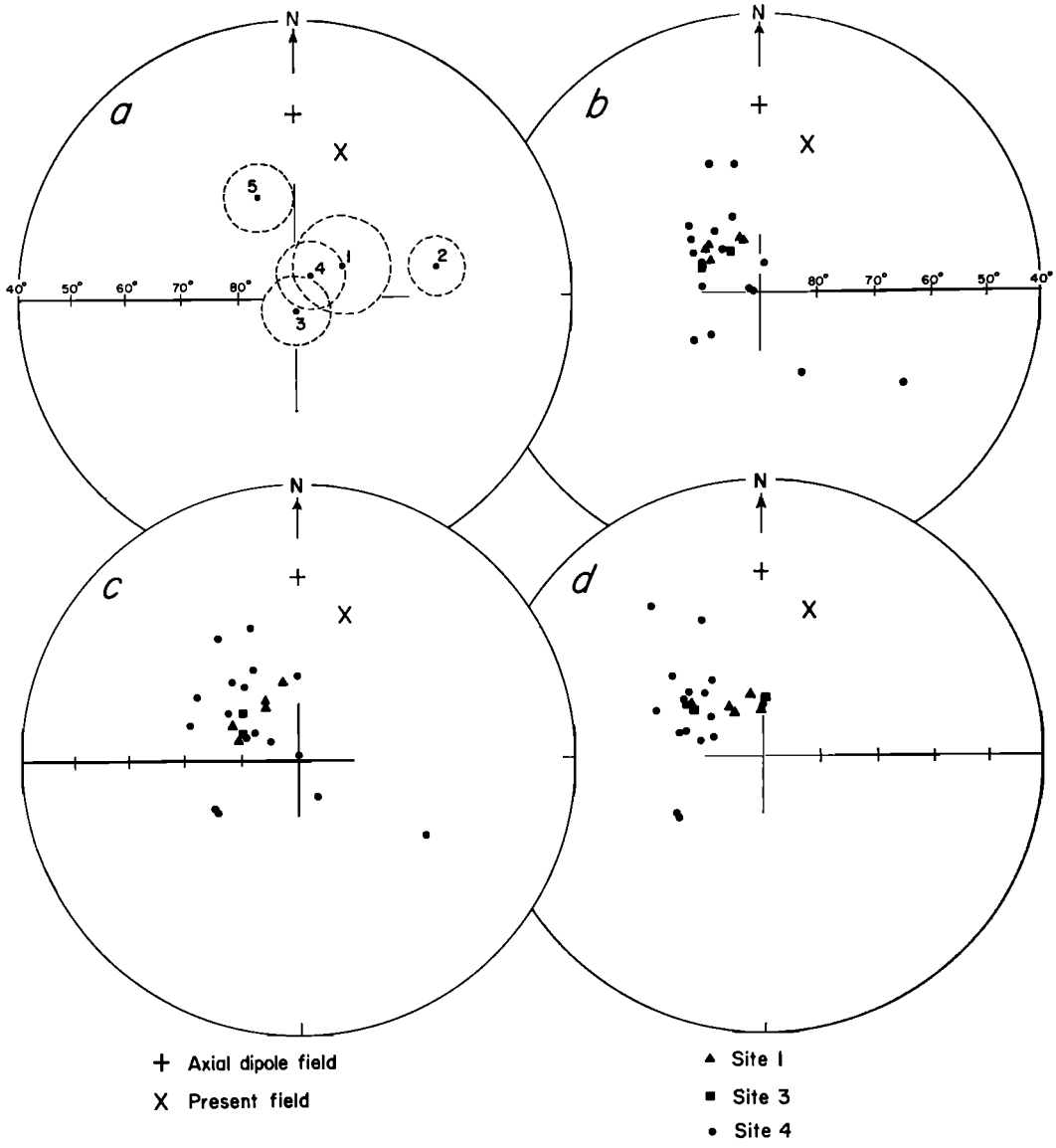


Fig. 4. Equal-area projections on lower hemisphere of north-seeking magnetization directions from sites in the Sonora Pass area. The inner 50° only of each projection is shown. (*a*) Mean directions at all sites, with circles of 95% confidence; original measurements (Table 4). (*b*, *c*, *d*) Directions in individual specimens with intensity of NRM greater than 7×10^{-4} emu/cc as first measured: (*b*) original measurements, (*c*) remeasurements, (*d*) after ac demagnetization in 200-oe peak field.

cussed by *Cox* [1964]. In this model the random magnetization is the sum of a number of random vectors all with the same intensity, resulting in a nearly Fisherian distribution.

In calculating the confidence interval for a mean pole, we have considered only the dispersion of the virtual poles with respect to the mean, and the individual uncertainties in virtual poles, i.e. their ovals of confidence, have been ignored. There is yet no rigorous way of including these, and the only alternative is the two-level scheme of *Watson and Irving* [1957]. The latter is not used here because the confidence interval that it gives applies only to the over-all

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TABLE 5. Average Values of Intensity of Natural Remanence J_n (as First Measured) and Susceptibility χ, for All Specimens of Granite and Autoliths

	Granites			Autoliths		
Site	N	$J_n \times 10^4$	$\chi \times 10^3$	N	$J_n \times 10^4$	$\chi \times 10^3$
A′	4	5.6	3.7	17	8.9	3.8
F′	7	1.4	1.7	9	2.8	1.7
G′	16	1.0	1.1	9	1.5	1.5
1	10	2.4	2.0	19	4.7	1.7
2	6	2.9	2.1	9	1.9	1.6
3	10	3.6	1.8	10	5.9	1.5
4	17	3.1	2.6	34	28.5	3.2
5	14	2.0	2.4	16	2.9	3.1

mean of all specimen measurements, and the fact that N ranges from 5 to 26 would in our opinion result in undue biasing of the over-all mean toward the sites with higher N. The procedure we have used, which appears to be customary in paleomagnetism, apparently results in an estimated α_{95} for a mean pole which is somewhat too small.

Ferromagnetic minerals. The magnetic characteristics of these granitic rocks and associated autoliths may be summarized as follows:

1. All specimens possess remanent magnetization, usually of intensity greater than 10^{-4} emu/cc and seldom less than 10^{-5} emu/cc.

2. The ratio of remanent magnetization to magnetization induced by the earth's field (the Koenigsberger ratio Q) is less than unity for all

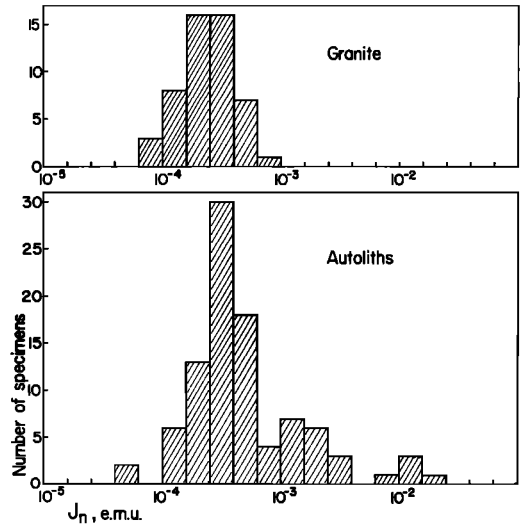


Fig. 5. Histograms of intensities of NRM as first measured in all specimens of granite and autoliths from the Sonora Pass area. Log of class interval = 0.2.

granites and autoliths from the Yosemite area and for all granites and most autoliths from the Sonora Pass area. These ratios are low in comparison with those of other igneous rocks, e.g. many basic plutonic rocks and most lavas.

3. Many specimens have a relatively large soft component of remanent magnetization, which might be described as viscous magnetization with a time constant of the order of days or weeks; this component is dominant in 85% of the specimens from the Sonora Pass area.

TABLE 6. NRM of Autolith Specimens from Sonora Pass Area Having Intensities Greater than 7×10^{-4} emu, with Virtual Magnetic South Poles
(*a:* First measured, August 1963. *b:* Remeasured, February 1964. *c:* After partial demagnetization in 200 oe peak alternating field.)

			Site-Mean					Virtual Pole			
Site	N		k	α_{95}	D	I	$J_n \times 10^4$	W. Long.	N. Lat.	δ_p	δ_m
1	5	*a*	483	3.5	−39	80	11.6				
		b	232	5.0	−40	79	12.5				
		c	276	4.6	−30	80	3.8	136	54	8.5	8.9
3	2	*a*	434	12.	−52	80	10.4				
		b	965	8.1	−56	78	10.9				
		c	82	28.	−32	79	3.2				
4	17	*a*	32	6.4	−50	84	53.2				
		b	32	6.4	−51	80	52.9				
		c	61	4.6	−57	75	11.2	156	49	7.7	8.5

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4. There are no correlations, direct or inverse, of susceptibility with intensity or stability of NRM.

5. In the Sonora Pass area, stability and consistency of magnetization directions are associated with anomalously high intensity of remanence. This is not true in the Yosemite area, where extremely divergent (but stable) magnetization directions are often associated with high intensities.

6. There is a general tendency for autoliths to be somewhat more strongly magnetized than the enclosing granitic rocks (even disregarding the anomalously strong autoliths in the Sonora Pass area), but there is no systematic difference in susceptibilities.

In an effort to account for some of these characteristics, in particular the strong magnetizations in the Sonora Pass autoliths, the identity and composition of the ferromagnetic minerals were investigated in some detail. *Curie et al.* [1963] had concluded on the basis of X-ray diffraction of ferromagnetic concentrates and limited Curie temperature data that the ferromagnetic mineral was a very pure magnetite; the present work has confirmed this.

A representative group of specimens was polished and examined in reflected light at magnifications up to 800×. By far the most abundant opaque mineral in all specimens is magnetite, in which ilmenite lamellae are absent. Sulfides are sparse or absent. About half of the specimens contain very small amounts of 'primary' ilmenite or ilmenite-hematite lamellar intergrowth; the remainder have none. Secondary hematite occurs in several specimens as an alteration (oxidation) product around margins and along the octahedral planes of magnetite grains, especially smaller grains. This secondary hematite is absent in specimens from the Yosemite area but is present in more than half of the specimens from the Sonora Pass area and is common (i.e. occupies up to 10% of original magnetite) in specimens from site 5. The occurrence of secondary hematite is unrelated to rock type (granite or autolith) or intensity and/or stability of magnetization. At site 5 its abundance can be correlated with a slightly 'rusty' appearance of the rock and alteration of biotite; thus it appears to be the product of recent (postglacial) weathering. There is no evidence of any significant component of chemical remanent magnetization (CRM) associated with the secondary hematite. However, sphene is ubiquitous in these rocks; it commonly surrounds the magnetite grains and apparently represents loss of Ti by the magnetite during postconsolidation cooling of the rock, possibly below the Curie temperature range.

Semiquantitative analyses of magnetite grains in nine of the polished specimens were made using an electron probe micro-analyzer. The X-ray emission spectra were scanned through the range of K wavelengths of the first transition series, from titanium through nickel. Atomic weight percentages of elements other than iron were estimated from K_α peak heights relative to the scaled height of the FeK_α peak, assuming 72 wt.% Fe in the magnetite. Systematic correction was made for background (Fe continuous spectrum); none was made for absorption by the specimen. The accuracy of this method is of the order of ±50% of the amount found, but, with the exception of nickel, the relative precision within each analysis is about ±15% or better. In view of the results reported below, the much higher accuracy attainable through the use of standard samples is felt to be unnecessary for the present purpose.

The analytical results are given in Table 7; titanium, cobalt, and nickel were undetectable in all magnetite grains analyzed. Because the absence of titanium was unexpected, the approximate detection limits are given for each analysis in Table 7. Traces of vanadium and maganese were found in all magnetites analyzed, and chromium is present in some, being perhaps more abundant in those from the Yosemite area. It appears quite certain that the content of foreign cations in these magnetites is always less than 1%; the average is about 0.5%. It is also clear from Table 7 that there is no significant correlation of trace-element content with lithology, presence of secondary hematite, or intensity of magnetization.

The uniformly high degree of purity and homogeneity and the relatively large grain size of the magnetite in these granitic rocks very probably account for the low Q ratios and the large soft component of magnetization. *Syono and Ishikawa* [1963*a*, *b*] have shown that in the magnetite-ulvospinel solid solution series, the magnetostriction and magnetocrystalline anisotropy constants are strongly dependent on

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TABLE 7. Semi-Quantitative Trace-Element Content of Magnetite in Selected Granite and Autolith Specimens, Obtained with Electron Probe Microanalyzer

Sample	Rock Type	Hematite Present	NRM $> 7 \times 10^{-4}$	Atomic Wt. % Ti	V	Cr	Mn
			Sonora Pass				
3	Autolith	No	Yes	<.03	.30	<.03	.13
4	Granite	No	No	<.02	.17	.06	.17
8	Autolith	Yes	No	<.02	.26	.05	.12
16	Autolith	Yes	Yes	<.02	.26	<.02	.06
18	Autolith	No	No	<.02	.20	<.02	.06
			Yosemite				
57	Granite	No	No	<.02	.30	.10	.18
59	Autolith	No	No	<.02	.42	.08	.05
74	Autolith	No	Yes	<.02	.31	.15	.14
77	Granite	No	No	<.05	.14	<.05	.11

composition, and in particular that the magnetostriction increases with increasing Ti content; thus, following *Verhoogen*'s [1959] theory, one might expect some titanomagnetites to carry a harder TRM than pure magnetite, which is confirmed by the data of *Uyeda* [1958, p. 16]. Further, *Roquet* [1954] has shown that TRM of magnetite (dispersed in a nonmagnetic matrix) is inversely dependent on grain size, which also accounts for the tendency of autoliths to be somewhat more strongly magnetized without having correspondingly greater susceptibilities [*Curie et al.*, 1963]. Remaining unaccounted for are the very strongly and uniformly magnetized autoliths at three of the Sonora Pass sites and the fact that taken as a whole the Sonora Pass rocks are much less stably magnetized than those in the Yosemite area.

Susceptibility anisotropy. The mean directions from the stably magnetized autoliths at sites 1 and 4 in the Sonora Pass area are nearly parallel to the foliation observed in the rocks (compare Figures 3 and 4*d*). Because the ferromagnetic material in these rocks consists of multidomain grains of magnetite, any preferred crystallographic orientation, to a first approximation and in weak fields, will not produce a bulk magnetocrystalline anisotropy in the rock and thus will not affect the magnetization directions. However, preferred orientation of long axes of non-equant magnetite grains can affect magnetization directions because of the resulting bias in the self-demagnetizing field. This kind of fabric can be detected, and its effect on NRM directions estimated, by measurements of susceptibility anisotropy [*Uyeda et al.*, 1963].

To make these measurements, we used a commercial 1000-cps susceptibility bridge (Geophysical Specialties, Inc.). This bridge measures the change in mutual inductance produced by placing a specimen between two of a set of three coaxial coils. The coils are about 4.5 cm in diameter and are separated by about 4.5 cm, and the peak field is estimated to be about 1 oe. The apparatus was modified to allow the cylindrical specimen (2.5 cm long, 2.5 cm in diameter) to be inserted laterally between two of the coils, with its axis normal to the coil axis. This allows easy determination of susceptibility anisotropy in the plane normal to the specimen axis. If the volume susceptibility is of the order of 10^{-3}, an anisotropy as small as 1% can be detected, and the orientation of a 5% anisotropy can be found to an accuracy of $\pm 3°$. Owing to the proximity of the specimen to the coils, the anisotropy cannot be determined in other planes unless a spherical specimen is used.

A preliminary test consisted of taking measurements at 10° intervals from 0° to 350°, on a specimen from site F′. Following *Uyeda et al.* [1963], we call P the ratio of maximum to minimum apparent susceptibilities; P for this specimen was 1.14. After opposite pairs of measurements were averaged to eliminate any effect due to eccentricity of the specimen, the maximum deviation from the best-fitting ellipse was 8% of the difference between the long and short

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axes, which is equivalent to 0.9% of the average susceptibility.

A further test was made to try to circumvent the necessity for using spherical specimens. Sets of three cores with mutually orthogonal axes were cut from each of two hand specimens of granite from sites 1 and 2, respectively. Although the susceptibilities measured at 10° intervals were again close fits to ellipses, for neither hand specimen did the entire set of measurements on the three cores describe a single triaxial ellipsoid; in other words, the P factor is variable in a single hand specimen over distances of a few centimeters. However, in both specimens the directions of the apparent maxima and minima in all three cores agreed to within 10°; this was expected because the cores had been cut parallel and normal to the observed fabric in the specimens. In the specimen from site 1 the fabric is only faintly discernible and appears to be a lineation within a foliation; the susceptibility ellipsoid is triaxial with the maximum paralleling the lineation, and the maximum P is 1.21. The specimen from site 2 is far more typical of all four sites in this area because it has a distinct foliation and no visible lineation. The susceptibility maximum is clearly parallel to the foliation; for both cores in which the plane of measurement is normal to the foliation the values of P were 1.32, but for the third core P in the plane of the foliation was still 1.14. This discrepancy is serious, and plainly there is little to be gained by determining the entire susceptibility ellipsoids in single small specimens of these coarse-grained granites.

To try to get some usable information, we measured the apparent anisotropies of all specimens from site 4 in the planes normal to their axes. There is an unmistakable tendency for the apparent susceptibility maximum in every specimen to parallel the foliation. No significant differences in anisotropy occur between granite and autoliths or between autoliths having strong and weak NRM's. Apparent anisotropies in the strongly magnetized autoliths ranged from $P = 1.02$ to $P = 1.15$; the latter is probably close to the true mean value. Using equation 11 of *Uyeda et al.* [1963] and neglecting the Hopkinson effect (the magnitude of which is unknown for these rocks), we have

$$\tan \theta_{obs} = \tan \theta_{ex}/P^2 \qquad (1)$$

where θ_{obs} is the angle between the observed NRM (assumed here to be TRM) and the susceptibility maximum, and θ_{ex} is the angle between the magnetizing field and the susceptibility maximum. The quantity $(\theta_{ex} - \theta_{obs})$ is the deflection of TRM toward the susceptibility maximum, being maximum near $\theta = 45°$. For $P = 1.15$ and $\theta_{ex} = 45°$, the deflection is only 8°. Owing to several assumptions and approximations [*Uyeda et al.*, 1963], this relation can only give a very rough estimate, but since for the Sonora Pass rocks $\theta_{obs} = 0°$, θ_{ex} must also have been very small in view of the small observed anisotropy. It seems safe to conclude that the magnetic meridian was fortuitously nearly parallel to the plane of the foliation in these rocks during the time they became magnetized.

In the previous paper [*Currie et al.*, 1963] it was concluded that inhomogeneity and anisotropy contributed little or nothing to the scatter of NRM directions. As pointed out in that paper, there is no visible fabric in the granite at any of the Yosemite sites, so that no systematic bias of directions should be expected. To check this and to try to estimate the contribution of possible random anisotropy to the dispersion of NRM, we measured the apparent susceptibility anisotropies of all specimens from site C. No tendency toward regularity of orientation of apparent susceptibility maxima or minima was noted. The average apparent P was 1.06, and the maximum apparent P was 1.18. The average true P very likely does not exceed 1.10, and so from (1) the maximum average deviation due to random anisotropy can be estimated as not exceeding 5.4°. At site C ($k = 28$) the mean angular deviation of NRM directions is 12.7°, so that we can conclude that random anisotropy probably contributes no more than about half of the observed dispersion of NRM directions. The average of the mean angular deviations of NRM directions for all paleomagnetically useful sites in the Yosemite area is 11.4°; thus this conclusion appears to hold in general for these rocks.

Paleomagnetic pole positions. Figure 6 shows, on an equal-area map of the northern hemisphere, the mean pole from the 12 Yosemite sites (Table 2), marked 'Y' with its circle of 95% confidence, and the two virtual poles from Sonora Pass sites 1 and 4 (Table 6) with their

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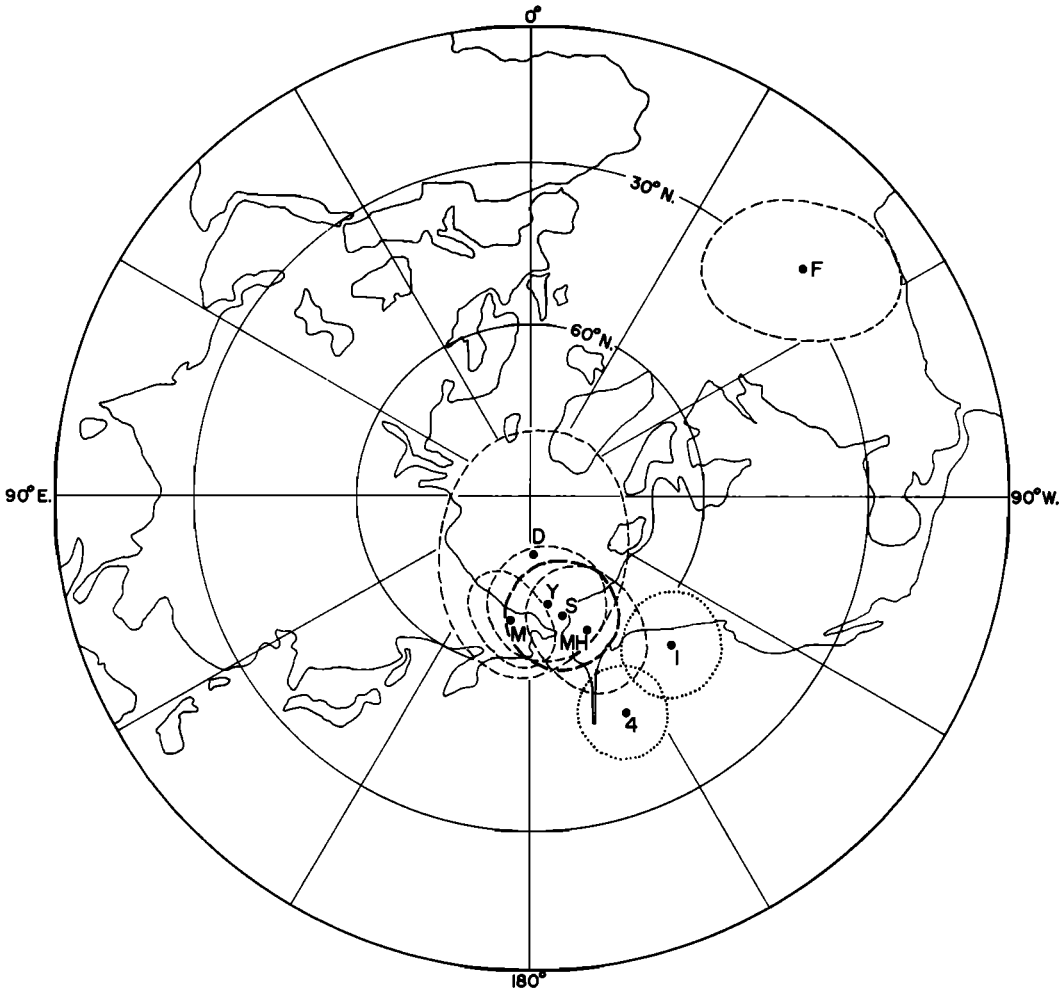


Fig. 6. Equal-area map of northern hemisphere showing all presently available paleomagnetic poles for the Cretaceous of North America, with 95% confidence ovals or circles. Y: mean of all virtual poles from Yosemite granites. 1, 4: virtual poles from Sonora Pass granites. S: mean of all virtual poles from Sierra Nevada granites. M, MH: Mt. Megantic and Monteregian Hills, respectively, in southeastern Quebec. D: Dakota sandstone, Arizona. F: Franciscan formation, California.

respective 95% confidence ovals. The positions of the Sonora Pass poles are not anomalous, in view of the dispersion of the Yosemite poles themselves (compare with poles G′ and H, Figure 2). All 14 virtual poles have been combined to give the mean pole marked 'S' in Figure 6, which is located at 164.8°W, 68.8°N ($k = 18$, $\alpha_{95} = 9.6°$); this pole is not very different from the pole obtained from the Yosemite data alone and is the most representative mean of the data from all the Sierra Nevada Cretaceous granites.

The only other paleomagnetic poles for the Cretaceous of North America for which radiometric ages are available have been obtained from intrusive igneous rocks in Quebec. The pole from Mt. Megantic [*Larochelle*, 1961] is shown in Figure 6 as 'M'; this single intrusion has been dated at 113 m.y. [*Fairbairn et al.*, 1963]. The mean pole from a group of alkaline intrusions, the Monteregian Hills [*Larochelle*, 1962], is shown in Figure 6 as 'MH'; these have been dated at 124 m.y. [*Fairbairn et al.*, 1963]. Neither of these poles is significantly different from the Sierra Nevada pole; that from Mt.

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Megantic diverges a little more but probably carries somewhat less weight since it only represents a single intrusion. The pole (H4 in the compilation by *Cox and Doell* [1960]) obtained from the Dakota sandstone in Arizona is marked 'D' in Figure 6; its 95% confidence oval encloses the poles already mentioned and also the geographic pole. Also shown in Figure 6 is an apparent pole position ('F') recently determined from Cretaceous or upper Jurassic lavas in the Franciscan formation of California, but for structural reasons the paleomagnetic significance of their magnetization is not clear [*Grommé and Gluskoter*, 1965].

Three closely coincident pole positions for North America have now been obtained from radiometrically dated igneous rocks spanning about 60 million years in the Cretaceous. The sampling sites are widely separated (Sierra Nevada, California; Monteregian Hills and Mt. Megantic, Quebec) and have been tectonically stable since the Cretaceous. The concordance of these paleomagnetic poles (1) supports our contention that the NRM in the Sierra Nevada granites is a true fossil magnetization and (2) suggests that these poles are representative of the geomagnetic pole position relative to North America during much of the Cretaceous. Preliminary results obtained in this laboratory from the 140-m.y.-old Valley Series batholiths mentioned in the introduction strongly support these conclusions (Grommé and Merrill, to be published). Pending completion of that work, we will not now make an extended comparison of the North American results with Cretaceous pole positions from other continents. It is interesting, however, that the Cretaceous poles derived from radiometrically dated Australian rocks [*Irving et al.*, 1963] are 45° (great-circle distance) away from the North American poles; this has already been pointed out by those authors with reference to earlier North American results.

Summary and conclusions. Re-evaluation of previously discarded data from three sampling sites in granites of the Yosemite area gives a pole, the mean of twelve site-poles, which differs very little from the previously published pole which was based on nine sites only. It is now clear that there is no systematic distribution of the individual site-poles, either with respect to apparent relative age or geographic position. Of 146 additional specimens collected from five sites in granites of the Sonora Pass area, which are of the same age (85 m.y.) as the Yosemite rocks, only 22 at two sites possessed NRM sufficiently stable to represent true paleomagnetic field directions. The mean pole calculated from these two sites and the twelve Yosemite sites again differs only little from the pole for the Yosemite rocks alone. It is located in the vicinity of northern Alaska and is not significantly different from poles obtained by *Larochelle* [1961, 1962] from 113- and 124-m.y.-old intrusive igneous rocks in southeastern Quebec.

The ferromagnetic mineral in all specimens examined is medium- to coarse-grained homogeneous magnetite which contains only about 0.5% of foreign cations. Ti was undetectable in all magnetite grains that were analyzed with the electron probe. This probably accounts for the relatively large and extremely soft component of magnetization which is manifested in the large dispersion of NRM directions at each site and in the failure of ac and thermal demagnetization to reduce the dispersion. About half of the observed angular dispersion appears to be due to random anisotropy. We cannot, however, account for the fact that the Sonora Pass samples were generally much less stably magnetized than the Yosemite samples, nor for the fact that some of the Sonora Pass samples were far more strongly and stably magnetized, in a uniform direction, than any of the others. The fundamental independent evidence for the true fossil nature of the NRM in these rocks is still its systematic divergence from the axial dipole and present field directions and its relative stability in the laboratory. From petrographic evidence, the magnetization appears to be some combination of TRM and a high-temperature CRM acquired below the Curie temperature during readjustment of the equilibrium composition of the magnetite with the simultaneous crystallization of sphene.

The agreement among the three independent radiometrically dated Cretaceous paleomagnetic poles is sufficiently close for us to suggest that they give further support to the assumption that is customarily made in paleomagnetism, namely that the geomagnetic field when averaged over sufficient time approximates a geocentric dipole. A time span of about 40 m.y. is represented by the three results mentioned; this

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may be an order of magnitude more time than is required for that approximation. It follows that if the average geocentric dipole has in fact always been nearly parallel to the earth's rotational axis, as it has been during the late Tertiary and Quaternary, then these three paleomagnetic poles may represent the approximate position of the geographic pole relative to North America, some 25° away from its present position. Nevertheless, the Cretaceous is one of the longest geological periods (about 70 m.y.), and many more well-dated paleomagnetic poles from tectonically stable areas in all continents are needed before the density of its paleomagnetic record will be comparable, for example, to those from the Permian or the Tertiary.

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